International Journal of Law and Psychiatry 55 (2017) 1–7

Contents lists available at ScienceDirect

International Journal of Law and Psychiatry

Public attitudes towards involuntary admission and treatment by mental health services in Norway

Inge Joa [a,b,*], Kjetil Hustoft [a], Liss Gøril Anda [a], Kolbjørn Brønnick [a,b], Olav Nielssen [c], Jan Olav Johannessen [a,b], Johannes H. Langeveld [a]

[a] Network for Clinical Psychosis Research, Division of Psychiatry, Stavanger University Hospital, Norway
[b] Network for Medical Sciences, Faculty of Social Sciences, University of Stavanger, Norway
[c] Faculty of Medicine and Health Sciences, Macquarie University, Sydney, New South Wales, Australia

ARTICLE INFO

Article history:
Received 19 December 2016
Received in revised form 16 June 2017
Accepted 13 September 2017
Available online 12 October 2017

Keywords:
Psychiatry
Involuntary hospitalization
Public opinion
Coercion

ABSTRACT

Purpose: The role of compulsory treatment of serious mental disorders has been the topic of ongoing public debate involving among others mental health professionals, service providers, service user advocates, relatives of service users, media commentators and politicians. However, relatively little is known about general public attitudes towards involuntary admission and compulsory treatment of people with various mental disorders. This article examines the attitudes in a representative sample of Norway's population towards the use of involuntary admission and treatment, and under which circumstances does the general public consider compulsory treatment to be justified in the Norwegian mental health care services.
Method: Data were collected from a representative sample of the population in Norway aged 18 and older. The sample was stratified for gender, geographical region and age distribution ($n = 2001$). The survey was performed in the months of May 2009 ($n = 1000$) and May 2011 ($n = 1001$), using Computer Assisted Telephone Interviews (CATI) by an independent polling company. All respondents were provided a general definition of coercive intervention before the interview was conducted.
Analysis: Univariate descriptions and bivariate analyses were performed by means of cross-tabulation, analysis of variance (one-way ANOVA) and comparing of group of means. Cohen's d was used as the measure for effect size.
Results: Between 87% and 97% of those surveyed expressed strong or partial agreement with the use of involuntary admissions or compulsory treatment related to specified cases and situations. The majority of interviewees (56%) expressed the opinion that overall, current levels are acceptable. A further, 34% were of the opinion that current levels are too low, while only 9.9% of respondents supported a reduction in the level of involuntary treatment. Lower levels of education were associated with a more positive attitude towards involuntary admission and treatment.
There was stronger support for admission to prevent suicide than the possibility of violence by the mentally ill.
Conclusion: The Norwegian adult population largely supports current legislation and practices regarding involuntary admission and compulsory treatment in the mental health services.
© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Most advanced countries have laws regulating involuntary admission and treatment of patients with serious mental disorders, and in most jurisdictions, involuntary admission also allows for involuntary treatment of the person's condition. However involuntary hospitalization is a controversial issue in psychiatry, due to the ethical complexity of admitting a person for treatment against their will. The threshold for use of compulsory treatment varies between countries, as do the rates of involuntary admissions, preventing a direct comparison of practices. The use of involuntary admission and coercive treatment (i.e., use of enforced medication, seclusion, use of restraint) is a topic of vigorous public debate in many countries, with mental health services often criticized for excessive use of such measures (Lartey, 2015). However, there have also been calls to increase the use of involuntary treatment, in order to protect the public from mentally ill individuals, and to ensure that people with mental illness who refuse treatment receive care (Brow, Shell, & Cole, 2015; Butler & Drakeford, 2005), and to prevent such individuals from harming themselves (Telegraph, 2015).

Complaints about involuntary admission have mainly come from service users and organizations representing patients, and from organizations working for the protection of civil liberties. However, support of

* Corresponding author at: Stavanger University Hospital, Division of Psychiatry, Postboks 8100, 4068 Stavanger, Norway.
E-mail address: ijo@sus.no (I. Joa).

https://doi.org/10.1016/j.ijlp.2017.09.002
0160-2527/© 2017 Elsevier Ltd. All rights reserved.

involuntary treatment has also been reported. For example, in a community survey conducted in France (Guedj, Sorum, & Mullet, 2012), 95% of participants agreed that involuntary admission is acceptable under certain conditions. Studies of service users' perspectives conducted in Ireland and Sweden also show that many patients accept the need for involuntary admission under certain circumstances. A German study (Angermeyer, Matschinger, & Schomerus, 2013) concluded that the general public almost invariably accepts compulsory admission if the person in question is perceived as violent. The majority of responders also supported involuntary admission in cases where a person threatens to commit suicide, or is unable to care for himself because of mental illness. In the study (Angermeyer, Matschinger, & Schomerus, 2014) the authors reported that >70% was principally for admitting persons to a hospital against their will, under certain conditions. When presented specific case-illustrations (violence, suicidal behavior) 80–95% of the population agreed to involuntary admission.

Previous Norwegian studies on attitudes to coercion report wide variation between the viewpoints of staff (Husum, Bjørngaard, Finset, & Ruud, 2011), the general public (Wynn, Myklebust, & Bratlid, 2006) and other stakeholders (Diseth, Bøgwald, & Høglend, 2011). Wynn et al. found that most members of the general public would permit use of coercion when the patients were violent, and between a third and a half supported coercive treatment when the patients had problems coping with activities of daily life, or when the patients were in the early phase of a psychotic illness, with few symptoms. No studies have found widespread opposition to the current law (Lauber & Rossler, 2007), which allows treatment for those deemed unable to care for themselves, or likely to deteriorate unless treated.

A review of published studies of attitudes to coercive treatment in the general population and among mental health professionals found support for involuntary treatment in most countries surveyed (Lauber & Rossler, 2007). Studies conducted in Ireland and Sweden found that patients also agree with the need for involuntary admission in certain circumstances (O'Donoghue et al., 2010; Wallsten, Ostman, Sjoberg, & Kjellin, 2008). Wallsten et al. interviewed patients admitted on both voluntary and involuntary basis, and their next-of-kin, and reported that "a majority of patients stated that it should be possible to compulsory admit patients", and "strong support for coercion in order to protect the patient and others was found among 'next-of-kins'".

In a study examining compulsory admission and treatment in schizophrenia in four European countries (England, Germany, Hungary and Switzerland) (Steinert, Lepping, Baranyai, Hoffmann, & Leherr, 2005), 1737 persons (psychiatrists, other psychiatric/medical personnel and laypeople) were asked about attitudes to compulsory admission and treatment. When asked whether they would agree with compulsory admission and compulsory treatment with antipsychotic medication based on typical clinical situations with refusal to consent to treatment, rates of agreement varied from 51 to 92% across countries and between 41 and 94% across professional groups. Interestingly psychologists and social workers in all countries presented a lower level of support for compulsory admissions and treatment in all countries than psychiatrists and laypeople. A further study by Lepping, Steinert, Gebhardt, and Röttgers (2004) compared attitudes about involuntary admission and treatment in mental health professionals and laypeople in England and Germany, and found that the attitudes of psychiatrists and other mental health workers were similar to those of the wider community in both countries. Furthermore, public attitudes towards psychiatric treatment appear to be evolving over time. A recently published systematic review of 162 population based surveys conducted between 2000 and 2015 (Angermeyer, van der Auwera, Carta, & Schomerus, 2017) showed that psychiatrists and psychologists were trusted to treat mental disorder, especially schizophrenia, and that the proportion of respondents who favored psychiatric hospital care and treatment with medication increased in the more recently conducted studies.

Involuntary admission and compulsory treatment in psychiatry raises important ethical and legal concerns. Legislation for coercive treatment for patients with severe mental disorders who reject voluntary treatment hinge on two principles: "capacity", i.e., a need for treatment coinciding with a loss of capacity to recognize that need, or "dangerousness", i.e., a conclusion that the person presents "a danger to self or others" because of mental illness, and cannot be cared for in a less restrictive way. In many countries, most notably the United States, the switch from capacity based mental health laws, to laws allowing compulsory treatment only when a person is deemed to be in danger of harming self or others was seen as a way of protecting patient's rights to refuse treatment. Lawsuits initiated by advocates of patient rights (O'Connor v. Donaldson, 1975) were interpreted to mean that dangerousness to self or to others was the only basis for involuntary care, despite the potential harm from non-treatment. However, mental health laws based on dangerousness alone, which apply in most states of the US, Australia and Canada, and in six out of thirty European countries, have the effect of limiting the availability of treatment to people who do not appear to be a danger to themselves or to others, thereby delaying the initial treatment of psychosis for nearly six months (Large & Nielssen, 2011). A delay of initial treatment of psychosis may in turn increase the risk of violent behavior (Large & Nielssen, 2011). The current Norwegian legislation covering involuntary admission and involuntary treatment has been developed gradually over a century and a half, based on the principle of solidarity with fellow citizens. The overall aim has been to ensure that clinical needs of those patients with severe mental illness who do not themselves recognize the need for treatment are still met, within a framework that protects their civil rights. The law not only covers the circumstances for involuntary admission, but also the conditions of care, the judicial review of patients in care and limits on the duration of detention, which protect the patient's rights and confirms the trust placed in services to provide care in the patients' best interests. The Norwegian Mental Health Care Act (Ministry of Health and Care Services, 1999) follows the principles of World Health Organization's (WHO, 2005) checklist which states that involuntary admittance and involuntary treatment; may only be given when A) there is evidence of a mental health disorder of specified severity; B) a serious likelihood that the person might do harm to self or others and/or substantial likelihood serious deterioration might occur in the patient's condition if treatment is not given, and; C) admission is for therapeutic purposes.

Media coverage of mental illness has been influential in shaping public understanding of mental illness and public attitudes towards involuntary treatment, especially among people with no personal experience of mental illness. In a Swedish study (Strömbäck, 2011) interviewing members of parliament and political journalists, most respondents felt that the media, rather than politicians, set the agenda and determine what issues are "important". News media stories often describe adverse events involving the mentally ill, but may also express editorial criticism of the paternalistic nature of services, or highlight scandals involving mental health services, such as the mistreatment of patients in hospitals. Consequently, Norwegian legislators (Authority, 2012) have come under pressure to limit the use of coercive treatment to remove the opportunity to involuntarily admit people at all, or alternatively, to abandon "capacity" criteria and to only retain "dangerousness" as a grounds for involuntary treatment.

The underlying presumption is that "the treatment principle" leads to a kind of overtreatment of people who could do well without, and that it represents a violation of basic human rights. In short "the treatment principle" means that it is possible to admit someone against their own will if there is a serious mental disorder present, such as psychosis, and there is the assumption that the patient will benefit from treatment. "The dangerousness principle" entails that one is allowed to admit someone against their own will if they represent a serious threat to somebody, in addition to having a psychosis.

To summarize, the development of legislation regarding coercive treatment in psychiatry is likely to be influenced by headlines and editorials in newspapers and other media, some of which urge for more use of coercive treatment, and others for more restrictions on involuntary care. In the absence of a large and well-conducted study of the attitudes of the general public towards involuntary psychiatric care, and the circumstances of that care, Norwegian legislators may perceive the opinions expressed in the media as being representative for the attitude of the general public.

The aim of this study was to provide a more detailed understanding of the attitude of the general public towards involuntary admission and treatment in psychiatric hospitals.

Based on the assumptions that the attitudes of the general public are

a) in disagreement with the current practice of mental health services and the existing legislation, and
b) in favor of reducing the use of involuntary treatment,

We posed the following research questions:

1. What are the attitudes of the general public towards the use of involuntary admission and treatment in the Norwegian mental health care services?
2. Under which circumstances does the general public consider compulsory treatment to be justified?

We formed the following hypotheses:

1. The use of involuntary admission and compulsory treatment in Norway exceeds the level considered acceptable by the general public.
2. The general public is in support of limiting the use of compulsory psychiatric hospital admission and treatment.

We also examined whether people with direct experience of caring for someone with mental illness are more in favor of involuntary treatment than are other members of the general public.

2. Methods

2.1. Subjects, design and analyses

Data were collected from a representative sample of the population in Norway aged 18 and older (Table 1). The sample was stratified for gender, geographical region and age distribution ($n = 2001$), margin of error ± 2–3%. The survey was performed in the months of May 2009 ($n = 1000$) and May 2011 ($n = 1001$), using Computer Assisted Telephone Interviews (CATI) by an independent polling company. All respondents were provided a general definition of coercive intervention before the interview was conducted.

Originally the study intended to explore changes in attitudes over time, but no changes were identified, and the two samples were as such handled as one group to increase the sample size. In addition to descriptive analyses, bivariate analyses were performed by means of cross-tabulation, analysis of variance (one-way ANOVA) and a comparison of group means. Cohen's d was calculated as a measure of effect size. Analyses were performed by use of SPSS (2014) ("SPSS Version 22.0").

Table 1
Background data all cases.

$N = 2001$		
Measure	Mean	SD
Age	50.8	17.5
Year education in addition to 10-year mandatory school	2.74	1.11
	N	Valid %
Male	*939*	*46.9*
Municipality population size		
<10,000	397	*19.8*
10,000 to 50,000	807	*40.3*
>50,000	797	*39.8*
Income (NOK)		
<400,000	490	*27.6*
400,000–799,000	705	*39.6*
>799,000	492	*27.7*
Missing	*313*	*15.7*
Employed	*1285*	*64.2*

Italic values indicates n.s (non-significant).

2.2. Instruments

Based on an earlier Norwegian survey on attitudes towards involuntary treatment developed by representatives from The Norwegian Psychiatrist Association, the Norwegian Psychologist Association, the Norwegian Nursing Association and the largest mental health consumers organization (Mental Helse), a 12 item questionnaire was developed, with 9 items consisting of statements regarding involuntary admission and compulsory treatment, scored on a 5-point Likert scale ranging from "completely agree" to "completely disagree". Out of these, 5 items asked for attitudes towards compulsory treatment of specific disorders or behaviors, while three were more general in nature.

The following 12 questions were asked:

1. To what degree are you confident that you would be getting the necessary health care if you were to develop a mental health disorder?
   (Five response categories, ranking from "very strong confidence" to "very little confidence".)
2. If someone in your close family were to develop a mental health disorder you could not handle yourself, who would you contact?
3. Is it your impression that there is too little, too much, or neither too much nor too little use of involuntary admissions and treatment in the mental health care services?
   (Items 4 through 11 with five response categories from "completely agree" to "completely disagree".)
4. Involuntary treatment in psychiatric mental health services is sometimes important to protect innocent people from danger and harm.
5. I would like for someone to take responsibility for me receiving effective treatment if I were to develop a serious mental disorder leaving me unable to take care of myself.
6. If someone with severe mental illness in your neighborhood was threatening/or frightening towards others, do you agree or disagree that he/she should be sent to a psychiatric examination even against his/her will?
7. If a family member were to express intent to commit suicide because of a serious depression, and I was unable to handle the situation on my own, I would take him/her to a psychiatric examination even against his/her will.
8. If one of your close relatives developed dementia do you agree or disagree that he/she should be sent to a psychiatric examination even against his/her will?
9. If someone close to me developed a serious mental disorder/psychiatric disorder, e.g. schizophrenia, and he/she refused to accept voluntary treatment, I would sign papers so that the person still got treatment.
10. Psychosis is a serious mental disorder with confusion or disturbed reality perception. If one of my next of kin had a psychosis and did not want to get treatment voluntarily, I would ensure they got this treatment involuntarily.
11. It is important that service users with bipolar disorder, who during manic periods may act uncritically and perform unreasonable actions like spending all their money, incur great debts, give away

Table 2
In the following cases, to what extent do you agree upon the use of involuntary admission or compulsory treatment?

$N = 2001$	Partly/completely agree combined (%)	Neither agree or disagree (%)	Partly or completely disagree (%)	Missing
To protect others against danger or violence	93.3	3.0	2.4	1.9
I would prefer for someone to act on my behalf If I were not able to take care of myself due to a serious psychiatric illness	96.9	1.8	1.3	1.1
If someone in your neighborhood threatens others	87.2	5.2	7.5	3.4
In case of depressive suicidal intention in a family member	95.0	1.5	3.4	2.2
In case of dementia	87.4	4.2	8.6	3.3
In case of psychosis	93.3	3.5	3.2	3.4
In case of bipolar disorder	90.5	4.5	5.0	3.1
In case of young people using drugs	91.7	3.8	4.5	1.9

fortunes, disgrace themselves or their family, or put themselves in great danger, should be taken care of and get treatment — even against their own will.

12. Young people using drugs like amphetamine and cannabis sometimes develop more severe psychotic disorders. These young people should be offered treatment even if they do not want such treatment.

2.3. Ethical review

The interviews were performed anonymously and according to The ICC/ESOMAR Code on Market and Social Research standards (ESOMAR). The only background information collected was demographic details (Table 1).

3. Results

In the participant sample, the distribution of gender, municipality population size and income were representative of the Norwegian population (Norway). Table 1 presents the sample demographics.

Depending on the nature of the severe mental health disorder described in survey items, between 87% and 97% of the respondents expressed strong or partial agreement to the use of involuntary admissions or compulsory treatment (Table 2) related to specific cases and situations (Items 1–12 in the questionnaire guide).

Responses to the items assessing attitudes towards level of involuntary admission and coercive treatment revealed that the majority of respondents (56%) expressed the opinion that in general, current levels are acceptable. A further 34% were of the opinion that current levels are too low, while 9.9% felt that the use of involuntary treatment should be reduced. In the sample, 31.5% reported a high level of confidence and trust that they would obtain adequate treatment in the mental health care system in the event of personally experiencing a severe mental health disorder, while 27.8% reported a low or very low confidence. The remaining 40.7% of subjects expressed a neutral attitude. When asked for their preferred point of contact with mental health services in the event of a severe mental disorder in a close relative, 71.9% said that they would seek help from their general practitioner, while 9.2% would contact the Emergency Unit, 6.4% would turn

Table 3
Effects of demographic variables on attitudes towards involuntary admission and treatment.

$N = 2001$	Too much coercive treatment $n = 137$	Too little coercive treatment $n = 470$	Neither too much or too little $n = 774$	Missing data (% of total)	F	Signif. (*p*-value, 2-sided)	Cohen's d	Odds ratio
Age (SD)	52.1 (17.00)	48.7 (16.24)	50.4 (17.96)	620 (31)	2.82 (df = 2)	0.06	0.127	1.26
	n (valid %)	(n) (valid %)	(n) (valid%)		$\times^2$		Cohen's d	Odds ratio
Female gender	76 (55)	263 (56)	388 (50)	620 (31	4.74 (df = 2)	0.11	0.11	1.23
Level of education (4 categories)								
10-year compulsory school	30 (10)	92 (33)	161 (57)	620 (31)	29.46 (df = 6)	0.000	0.21	1.46
1–3 year high school	12 (6)	94 (46)	100 (48)					
4–6 year college/university/high school	40 (9)	170 (37)	247 (54)					
>6 years of university level education	55 (13)	114 (826)	266 (61)					
Employed (yes/no)	82 (60)	327 (70)	500 (65)	620 (31)	5.625 (df = 2)	0.060	0.13	1.26
Income (3 categories) annual salary NOK								
<400,000	36 (11)	121 (36)	180 (53)		(df = 6) 6.874	0.333	0.11	1.21
400,000–799,000	37 (7.5)	174 (35)	284 (57)	770 (38.5)				
>799,000	40 (11)	116 (33)	195 (56)					
Community size (3 categories) Community size, population								
<10,000	25 (10)	94 (37)	138 (54)		(df = 4) 1.351	0.853	0.04	1.08
10,000 to 50,000	53 (9)	191 (34)	320 (57)	620 (31)				
>50,000	59 (11)	185 (33)	316 (56)					
Own experience of the care of someone with mental health problems	48 (59)	127 (61)	213 (52)	1302 (65)	(df = 2) 5.58	0.06	0.018	1.38

Chi-Square Tests

	Value	df	Asymptotic significance (2-sided)
Pearson Chi-Square	29,456[a]	6	,000
Likelihood ratio	29,723	6	,000
Linear-by-linear association	4683	1	,030
N of valid cases	1381		

[a] 0 cells (0,0%) have expected count less than 5. The minimum expected count is 20,44.

to their local hospital, 5.4% to a psychologist, 1.5% to the police, and 5.4% to others.

Further, we tested for possible effects of demographic variables on attitudes to involuntary admission and treatment (Table 3). Respondents with lower levels of education demonstrated a more positive attitude towards involuntary admission and treatment than did subjects with higher education. We found no significant differences in attitudes towards involuntary admission and treatment for variables of age, gender, employment status, income or size of local community living, nor did we find any significant effect of the direct experience of the care of someone with mental health problems on attitudes towards involuntary admission and treatment.

Finally, we compared the respondents' attitudes towards the use of coercive treatment in different scenarios where use of involuntary admission might be an option (Table 4). We found that more respondents would favor involuntary admittance if a family member expressed suicidal ideation, than if a neighbor with severe mental illness was to threaten others. The ranking of scenarios in order of greatest proportion of respondents in disfavor of involuntary coercion was as follows: 1) If someone in your neighborhood with severe mental illness threatens others, and In case of dementia; 2) In case of bipolar disorder; 3) In case of psychosis, and In case of young people using drugs; 4) In case of depressive suicidal intention in a family member.

4. Discussion

The main finding of this study of the attitudes of a representative sample of the adult Norwegian population was the public support for current levels of involuntary admission and treatment. This support is most firm in the cases of psychosis, and severe depression with suicidal ideation. Hence, we did not find support for our first hypothesis that the Norwegian public is in favor of curtailing the use of coercive psychiatric treatment. This finding is surprising given the generally negative attitudes expressed in the Norwegian media towards involuntary treatment, which has mostly been against the use of compulsory treatment (Anda, 2013). Our results show that the views of the media on this subject do not necessarily represent the views of the wider community. Further, the critical comments in the media about psychiatric care do not appear to have turned public opinion against the use of involuntary treatment. An interesting finding was that personal experience with psychiatric services had no effect on attitudes towards coercive treatment. About half of our sample reported some experience with psychiatric services, either first hand as a service user, or second hand through the illness of a relative, employee or elsewhere in their social network, but familiarity with psychiatric treatment was not associated within a change in attitudes towards coercive treatment in either direction. Media is a main source for general information to the public. In recent years attempts to reduce the stigma related to mental illness by media may have played a significant role. This is supported in a UK study by (Goulden et al., 2011). In a sample of 1361 newspaper articles, the authors (Goulden et al., 2011) found a significant proportional reduction in articles covering negative aspects about mental illness between 1992 and 2008. In the same period there was a significant increase in articles focusing on educative aspects about psychiatric disorders. The study reported that the coverage improved for depression but remained negative for schizophrenia. As such one might assume that the stigma issues connected to mental disorders and involuntary measures are complex and important. On the one hand, too much use of involuntary approaches can contribute to increased stigma and self-stigma (self-stigma occurs when people internalize public prejudice and discrimination and suffer numerous negative consequences as a result; Corrigan & Rao, 2012), while on the other hand too little can contribute to a public perception that people with serious mental disorders such as schizophrenia are much more dangerous than the "normal" population and that the level of involuntary treatment is too low.

In contrast with our expectation, we found a stronger public support for coercion based on perceived dangerousness to self or to others by caregivers. The strongest support for involuntary treatment was in cases where the recipient was unable to take care of him or herself, for example, in cases of depressive illness with suicidal ideation. Surprisingly, and in contrast to previous studies, involuntary admission and treatment of potentially violent mentally ill individuals did not attract the same level of support in the community. This may be because public experience with dangerous acts or threatening behavior in mentally ill people is rare, whereas many people have observed the effects of untreated mental illness on self-care. The firm support for the treatment principle in compulsory care might also be related to aspects of the Norwegian culture and health care system. Social democratic ideology has guided post-war Norwegian politics and is embedded in the concept of the Welfare State including a comprehensive and highly trusted public health system. The medical profession in particular is highly regarded and is generally believed to act in the best interest of those in need for health care, including psychiatric care. People might also simply have

Table 4
Comparison of attitudes towards coercive treatment regarding different scenarios in which coercive treatment can be performed.

$N = 2001$ Test of significance based on comparison of z-scores $p = 0.05$ (t-test)	Mean agreement score (sd)	If someone in your neighborhood with severe mental illness threatens others	In case of depressive suicidal intention in a family member	In case of dementia	In case of psychosis	In case of bipolar disorder	In case of young people using drugs
Mean agreement score (sd)		1.60 (0.981)	1.30 (0.718)	1.61 (1.049)	1.40 (0.746)	1.50 (0.851)	1.42 (0.830)
If someone with severe mental illness in your neighborhood was to threatens others	1.60 (0.981)	X	12.814 (1908) 0.000*	−0.189 (1879) 0.850*	8.480 (1879) 0.000*	4.003 (1885) 0.000*	7.538 (1908) 0.000*
In case of depressive suicidal intention in a family member	1.30 (0.718)		X	−12.714 (1901) 0.000*	−5.077 (1905) 0.000*	−9.212 (1909) 0.000*	−5.539 (1927) 0.000*
In case of dementia	1.61 (1.049)			X	8.530 (1884) 0.000*	4.037 (1888) 0.000*	7.398 (1908) 0.000*
In case of psychosis	1.40 (0.746)				X	−4.700 (1893) 0.000*	−0.834 (1913) 0.404*
In case of bipolar disorder	1.50 (0.851)					X	4.159 (1915) 0.000*
In case of young people using drugs	1.42 (0.830)						X

Agreement categories 1: completely agree; 2: partly agree; 3: neither agree or disagree; 4: partly disagree; 5: completely disagree; 6: do not know (coded as missing).
* T-values, (degrees of freedom), p-value.

disagreed with the premise that the odd and frightening behavior of the neighbor described in our survey vignette is sufficient grounds for compulsory hospital admission, even though they agree that compulsory treatment is acceptable in principle.

Our findings show that the general public does not see a need to further limit the current level of involuntary admission and treatment, and that the call for more restrictive legislation regarding the use of involuntary care in certain sections of the media does not reflect public opinion. The Norwegian government (Ministry of Health and Care Services, 2013) has encouraged psychiatric health trusts to gradually reduce involuntary admission. However, it was not clear whether this target was set to encourage health services to do a better job in preventing the need for involuntary admission through earlier intervention or improved continuity of care, or if it was set as a way of measuring improvements in the protection of human rights of patients. The target did not appear to be based on any scientific evidence, and with our findings suggesting that it is not backed by the demands of the electorate, instead it appears to be based on perceived ethical or political considerations.

Involuntary admissions and compulsory treatment encompass legal and moral issues. Coercive treatment should be kept to the minimum required for optimal treatment, and mental health professionals should always aim to minimize the level of coercive treatment. Early intervention results in a better prognosis and higher degree of functional recovery in schizophrenia and other psychoses (Hegelstad et al., 2012; Husum et al., 2011). Given that delayed treatment of severe mental illness might over time lead to a greater need for coercive treatment (Large & Nielssen, 2011), a broader investment in early intervention and treatment, and improved continuity of care might lead to the desired reduction of compulsory measures, at the same time safeguarding the human rights of mental health service users.

Stricter regulations that limit the circumstances under which compulsory admission and treatment may be used, might appear to be a way to reduce the use of such treatment. However, raising the threshold for involuntary treatment has had unintended and even disastrous consequences elsewhere (Appelbaum, 1997; Large, Nielssen, & Lackersteen, 2009; Large, Nielssen, Ryan, & Hayes, 2008), including an increase in violence by the mentally ill, and even a paradoxical increase in involuntary care. Another means for reaching the goal of reduced involuntary admission and treatment might be the implementation of Psychiatric Advance Directives (Zelle, Kemp, & Bonnie, 2015). These allow adults with decision-making capacity to declare their treatment preferences and/or to designate proxy decision makers to act on their behalf should they be deemed incapable in the future of making informed choices on their own. However, the development of such directives might be hampered by the legislator's limited knowledge about more severe mental health problems. This mental health literacy might be less than required to develop such directives both among legislators as well as the actual individual (Brown, 2003; Lloyd & King, 2003). Regulation that limits the circumstances in which compulsory admission and treatment may be used might also appear to be a way to reduce the use of compulsory treatment.

4.1. Strengths and limitations

4.1.1. Strengths

The methods employed by the study, which surveyed independent stratified samples representative of the Norwegian general public, using unbiased interviewers who did not have any specific opinion on the subject.

4.1.2. Limitations

No information on the numbers and characteristics of people who refused to participate in the survey, who may have been less compliant to authority, are available. The survey was administered in 2009 and 2011. However, there have been no changes in legislation since then, and the demographic distribution of the Norwegian population has not changed significantly, so we consider that the attitudes of our samples remain representative for the Norwegian population.

Despite our best efforts to formulate the items using neutral language, the nature of the questions might themselves have been leading. The questions did not attempt to establish whether people distinguished in their attitudes between involuntary admission on the one hand, and coercive administration of medication or physical restraint on the other.

5. Conclusion

The findings of this study demonstrate widespread public support for the current system of psychiatric care and for the retention of the current provisions for involuntary care in Norway.

Acknowledgements

We acknowledge the assistance of Thore Gaard Olaussen, Respons Analyse AS, Bredalsmarken 15, 5006 Bergen, Norway.

The survey in 2009 was performed by the Norwegian Psychiatric Association in cooperation with the Norwegian Psychological Association and the Norwegian Association for Mental Health Nurses.

Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

Anda, L. G. (2013). Mord, galskap og løver – schizofreni i norske aviser. *Tidsskrift for Norsk Psykologforening*, *50*(12), 1154–1162 (50(12), 8).

Angermeyer, M. C., Matschinger, H., & Schomerus, G. (2013). Attitudes towards psychiatric treatment and people with mental illness: Changes over two decades. *Br. J. Psychiatry*, *203*, 146–151.

Angermeyer, M. C., Matschinger, H., & Schomerus, G. (2014). Attitudes of the German public to restrictions on persons with mental illness in 1993 and 2011. *Epidemiology and Psychiatric Sciences*, *23*(3), 263–270. https://doi.org/10.1017/S2045796014000183.

Angermeyer, M. C., van der Auwera, S., Carta, M. G., & Schomerus, G. (2017). Public attitudes towards psychiatry and psychiatric treatment at the beginning of the 21st century: A systematic review and meta-analysis of population surveys. *World Psychiatry*, *16*(1), 50–61.

Appelbaum, P. S. (1997). Almost a revolution: An international perspective on the law of involuntary commitment. *The Journal of the American Academy of Psychiatry and the Law*, *25*(2), 135–147.

Brow, J., Shell, Y., & Cole, T. (2015). *Forensic psychiatry, theory, research, policy and practice*. London: Sage.

Brown, M. (2003). The law and practice associated with advance directives in Canada and Australia: Similarities, differences and debates. *Journal of Law and Medicine*, *11*(1), 59–76.

Butler, P. I., & Drakeford, M. (2005). *Scandal, social policy and social welfare* (2nd ed.). Bristol: Policy Press.

Corrigan, P. W., & Rao, D. (2012). On the self-stigma of mental illness: Stages, disclosure, and strategies for change. *Canadian Journal of Psychiatry. Revue Canadienne de Psychiatrie*, *57*(8), 464–469.

Diseth, R. R., Bøgwald, K. -P., & Høglend, P. A. (2011). Attitudes among stakeholders towards compulsory mental health care in Norway. *International Journal of Law and Psychiatry*, *34*(1), 1. https://doi.org/10.1016/j.ijlp.2010.11.001.

ESOMAR (d). ESOMAR codes and guidelines are available. From www.esomar.org.

Goulden, R., Corker, E., Evans-Lacko, S., Rose, D., Thornicroft, G., & Henderson, C. (2011). Newspaper coverage of mental illness in the UK, 1992–2008. *BMC Public Health*, *11*, 796. https://doi.org/10.1186/1471-2458-11-796.

Guedj, M., Sorum, P. C., & Mullet, E. (2012). French lay people's views regarding the acceptability of involuntary hospitalization of patients suffering from psychiatric illness. *International Journal of Law and Psychiatry*, *35*(1), 50–56. https://doi.org/10.1016/j.ijlp.2011.11.010.

Hegelstad, W. t. V., Larsen, T. K., Auestad, B., Evensen, J., Haahr, U., Joa, I., ... McGlashan, T. (2012). Long-term follow-up of the TIPS early detection in psychosis study: Effects on

10-year outcome. *American Journal of Psychiatry*, *169*(4), 374. https://doi.org/10.1176/appi.ajp.2011.11030459 ((Author abstract) (Report)).

Husum, T., Bjørngaard, J., Finset, A., & Ruud, T. (2011). Staff attitudes and thoughts about the use of coercion in acute psychiatric wards. *The International Journal for Research in Social and Genetic Epidemiology and Mental Health Services*, *46*(9), 893–901. https://doi.org/10.1007/s00127-010-0259-2.

Large, M. M., & Nielssen, O. (2011). Violence in first-episode psychosis: A systematic review and meta-analysis. *Schizophrenia Research*, *125*(2–3), 209–220. https://doi.org/10.1016/j.schres.2010.11.026.

Large, M. M., Nielssen, O. B., & Lackersteen, S. M. (2009). Did the introduction of 'dangerousness' and 'risk of harm' criteria in mental health laws increase the incidence of suicide in the United States of America? *Social Psychiatry and Psychiatric Epidemiology*, *44*(8), 614–621. https://doi.org/10.1007/s00127-008-0488-9.

Large, M. M., Nielssen, O., Ryan, C. J., & Hayes, R. (2008). Mental health laws that require dangerousness for involuntary admission may delay the initial treatment of schizophrenia. *Social Psychiatry and Psychiatric Epidemiology*, *43*(3), 251–256. https://doi.org/10.1007/s00127-007-0287-8.

Lartey, J. (2015). *Natasha McKenna's death shows the need for restraint — Not restraints.* theGuardian Retrieved from http://www.theguardian.com/commentisfree/2015/sep/11/natasha-mckenna-video-death-restraint.

Lauber, C., & Rossler, W. (2007). Involuntary admission and the attitude of the general population, and mental health professionals. *Psychiatrische Praxis*, *34*(Suppl. 2), S181–185. https://doi.org/10.1055/s-2006-952,002.

Lepping, P., Steinert, T., Gebhardt, R. -P., & Röttgers, H. R. (2004). Attitudes of mental health professionals and lay-people towards involuntary admission and treatment in England and Germany—A questionnaire analysis. *European Psychiatry*, *19*(2), 91–95. https://doi.org/10.1016/j.eurpsy.2003.11.001.

Lloyd, C., & King, R. (2003). Consumer and carer participation in mental health services. *Australasian Psychiatry*, *11*(2), 180–184.

Ministry of Health and Care Services (1999). *Norwegian Mental Health Act revised June 2nd 1999 no 62. Lov om etablering og gjennomføring av. psykisk helsevern av. 2. juli 1999 nr 62.*

Ministry of Health and Care Services (2013). *Mission document for The Western Norway Regional Health Authority.*

Norwegian Directorate of Health (2012). *Annual report 2011.* (Oslo).

O'Connor v. Donaldson (1975). *422 U.S. 563.*

O'Donoghue, B., Lyne, J., Hill, M., Larkin, C., Feeney, L., & O'Callaghan, E. (2010). Involuntary admission from the patients' perspective. *Social Psychiatry and Psychiatric Epidemiology*, *45*(6), 631–638. https://doi.org/10.1007/s00127-009-0104-7.

SPSS (2014). *Version 22.0 (Chicago, IL, USA).*

Statistics Norway (d). from https://www.ssb.no/en/inntekt-og-forbruk/statistikker/ifhus/aar.

Steinert, T., Lepping, P., Baranyai, R., Hoffmann, M., & Leherr, H. (2005). Compulsory admission and treatment in schizophrenia. *Social Psychiatry and Psychiatric Epidemiology*, *40*(8), 635–641. https://doi.org/10.1007/s00127-005-0929-7.

Strömbäck, J. (2011). Mediatization and perceptions of the media's political influence. *Journalism Studies*, *12*(4), 423–439. https://doi.org/10.1080/1461670X.2010.523583.

Telegraph, T. (2015). *Mother who lay in front of train with baby son had been told psychiatrist could not help, inquest hears.* The Telegraph Retrieved from http://www.telegraph.co.uk/news/uknews/law-and-order/11507669/Mother-who-lay-in-front-of-train-with-baby-son-had-been-told-psychiatrist-could-not-help-inquest-hears.html.

Wallsten, T., Ostman, M., Sjoberg, R. L., & Kjellin, L. (2008). Patients' and next-of-kins' attitudes towards compulsory psychiatric care. *Nordic Journal of Psychiatry*, *62*(6), 444–449. https://doi.org/10.1080/08039480801984248.

WHO. (2005). WHO resource book on mental health, human rights and legislation. (vol. 199): World Health Organization.

Wynn, R., Myklebust, L. -H., & Bratlid, T. (2006). Attitudes to coercion among health-care workers and the general public in Norway. *Journal of Psychiatric Intensive Care*, *2*(1), 31–37. https://doi.org/10.1017/S1742646406000252.

Zelle, H., Kemp, K., & Bonnie, R. J. (2015). Advance directives in mental health care: Evidence, challenges and promise. *World Psychiatry*, *14*(3), 278–280. https://doi.org/10.1002/wps.20268.